

Overview of Autoregressive model

Subject: Autoregressive model

1.

Estimation of AR parameters The above equations (the Yule–Walker equations) provide several routes to estimating the parameters of an AR(p) model, by replacing the theoretical covariances with estimated values. Some of these variants can be described as follows:

2.

Calculation of the AR parameters There are many ways to estimate the coefficients, such as the ordinary least squares procedure or method of moments (through Yule–Walker equations). The AR(p) model is given by the equation

3.

have been estimated, the autoregression can be used to forecast an arbitrary number of periods into the future. First use t to refer to the first period for which data is not yet available; substitute the known preceding values X_{t-i} for $i=1, \dots, p$ into the autoregressive equation while setting the error term ε_t equal to zero (because we forecast X_t to equal its expected value, and the expected value of the unobserved error term is zero). The output of the autoregressive equation is the forecast for the first unobserved period. Next, use t to refer to the next period for which data is not yet available; again the autoregressive equation is used to make the forecast, with one difference: the value of X one period prior to the one now being forecast is not known, so its expected value—the predicted value arising from the previous forecasting step—is used instead. Then for future periods the same procedure is used, each time using one more forecast value on the right side of the predictive equation until, after p predictions, all p right-side values are predicted values from preceding steps. There are four sources of uncertainty regarding predictions obtained in this manner: (1) uncertainty as to whether the autoregressive model is the correct model; (2) uncertainty about the accuracy of the forecasted values that are used as lagged values in the right side of the autoregressive equation; (3) uncertainty about the true values of the autoregressive coefficients; and (4) uncertainty about the value of the error term ε_t for the period being predicted. Each of the last three can be quantified and combined to give a confidence interval for the n -step-ahead predictions; the confidence interval will become wider as n increases because of the use of an increasing number of estimated values for the right-side variables.

4.

It can be seen that the autocovariance function decays with a decay time (also called time constant) of $\tau = 1 / (1 - \phi)$. The spectral density function is the Fourier transform of the autocovariance function. In discrete terms this will be the discrete-time Fourier transform:

5.

Implementations in statistics packages R – the stats package includes ar function; the astsa package includes sarima function to fit various models including AR. MATLAB – the Econometrics Toolbox

and System Identification Toolbox include AR models. MATLAB and Octave – the TSA toolbox contains several estimation functions for uni-variate, multivariate, and adaptive AR models. PyMC3 – the Bayesian statistics and probabilistic programming framework supports AR models with p lags. bayesloop – supports parameter inference and model selection for the AR-1 process with time-varying parameters. Python – statsmodels.org hosts an AR model.

6.

$$S(f) = \frac{\sigma_Z^2}{1 + \varphi_1^2 + \varphi_2^2 - 2\varphi_1 \cos(2\pi f) - 2\varphi_2 \cos(4\pi f)} \quad \{ \displaystyle S(f) = \frac{\sigma_Z^2}{1 + \varphi_1^2 + \varphi_2^2 - 2\varphi_1 \cos(2\pi f) - 2\varphi_2 \cos(4\pi f)} \}$$

7.

Explicit mean/difference form of AR(1) process The AR(1) model is the discrete-time analogy of the continuous Ornstein-Uhlenbeck process. It is therefore sometimes useful to understand the properties of the AR(1) model cast in an equivalent form. In this form, the AR(1) model, with process parameter $\theta \in \mathbb{R}$, is given by

8.

which can be solved for all $\{\varphi_m; m = 1, 2, \dots, p\}$. The remaining equation for $m = 0$ is

9.

$$z_1, z_2 = \frac{1}{\sqrt{\varphi_1^2 + 4\varphi_2^2}} \left(\varphi_1 \pm \sqrt{\varphi_1^2 + 4\varphi_2^2} \right) \quad \{ \displaystyle z_1, z_2 = \frac{1}{\sqrt{\varphi_1^2 + 4\varphi_2^2}} \left(\varphi_1 \pm \sqrt{\varphi_1^2 + 4\varphi_2^2} \right) \}$$

10.

$$\Phi(\omega) = \frac{1}{\sqrt{2\pi}} \frac{\sigma_\varepsilon^2}{1 - \varphi_1^2} \frac{1}{\sqrt{1 - \varphi_1^2 + \varphi_2^2}} \quad \{ \displaystyle \Phi(\omega) = \frac{1}{\sqrt{2\pi}} \frac{\sigma_\varepsilon^2}{1 - \varphi_1^2} \frac{1}{\sqrt{1 - \varphi_1^2 + \varphi_2^2}} \}$$

11.

In statistics, an autoregressive (AR) model is a modelled representation of a type of random process. It can be used to describe time-varying processes from many natural and artificial sources. The model specifies output variables that are dependent linearly on their own previous values on a stochastic basis. The model is in the form of a stochastic difference equation (or recurrence relation) which should not be confused with a differential equation. Together with the moving-average (MA) model, it is a special case and key component of the more general autoregressive–moving-average (ARMA) and autoregressive integrated moving average (ARIMA) models of time series, which have a more complicated stochastic structure; it is also a special case of the vector autoregressive model (VAR), which consists of a system of more than one interlocking stochastic difference equation in more than one evolving random variable. Another important extension is the time-varying autoregressive (TVAR) model, where the autoregressive coefficients are allowed to change over time to model evolving or non-stationary processes. TVAR models are widely applied in cases where the underlying dynamics of the system are not constant, such as in sensors time series modelling, climate science, economics and

finance (as econometrics), signal processing, telecommunications, radar systems, and biological signals. Unlike the moving-average (MA) model, the autoregressive model is not always stationary; non-stationarity can arise either due to the presence of a unit root or due to time-varying model parameters, as in time-varying autoregressive models. Large language models are called autoregressive, but they are not a classical autoregressive model in this sense because they are not linear.